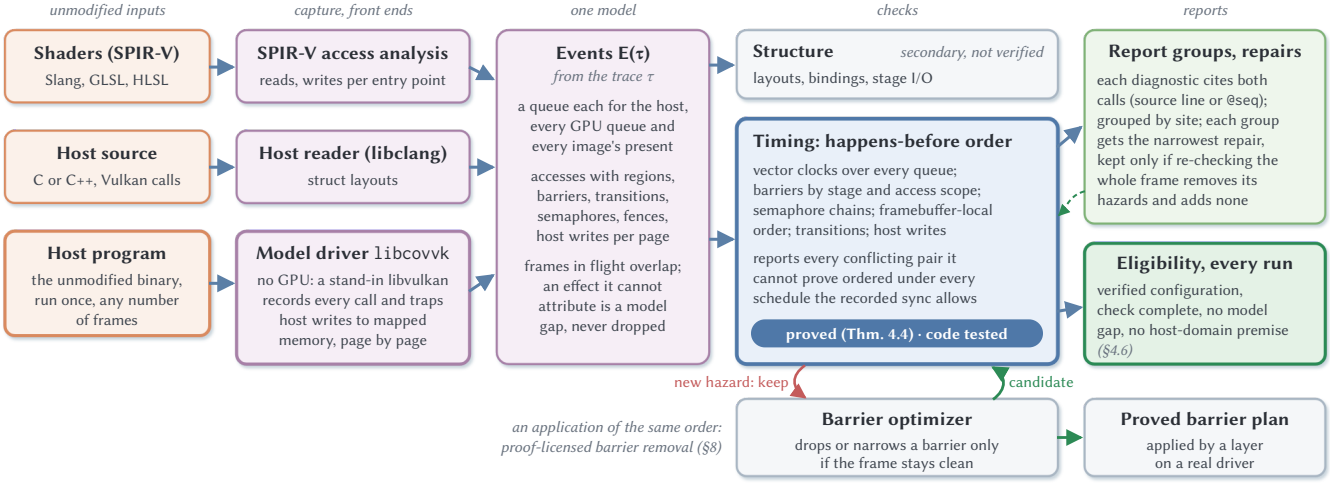


What a clean, eligible report rests on *each link is a premise, trusted and tested code, a proof, or a check on every run (Tables 2 and 4)*

1 Real execution
one run of the unmodified program; every call and host write recorded

premise about capture

2 Event extraction
trace and shaders become the events of Def. 4.1; gaps reported, not dropped

trusted, tested code; premises about the model

3 Abstract checker
reports every pair of conflicting accesses that the defined happens-before order leaves unordered

proved by hand (Thm. 4.4)

4 Eligibility
the run used the verified configuration, with no gap and no host-domain premise

checked per run; it cannot check links 1 and 2

No report ⇒ race-free
under every device schedule consistent with the recorded synchronization (Cor. 4.5), for the modelled accesses on the recorded host path, if the premises of 1 and 2 hold

Beyond the recorded frames: Theorem 4.11 extends a clean result to later frames of a program that repeats its period (premise P-PER, which a trace cannot establish).